

ROBERT CADDY

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Research Interest: Computational Methods in Astrophysics & Scientific Software Best Practices

EDUCATION

University of Pittsburgh, Pittsburgh, PA 2018-2024

Ph.D. - Physics

Dissertation Title: *Cholla-MHD: An Exascale-Capable Magnetohydrodynamic Extension to the Cholla Astrophysical Simulation Code*

Advisor: Dr. Evan Schneider

Bowling Green State University, Bowling Green, OH 2016-2018

M.S. - Physics

Thesis Title: *Time Series Photometry of the Symbiot Star V1835 Aql and New Variable Stars in Aquila*

Advisor: Dr. Andrew Layden

Purdue University, West Lafayette, IN 2012-2016

B.A. - Honors Physics, Astronomy minor

EMPLOYMENT & RESEARCH EXPERIENCE

Research Software Engineer II, Princeton University, Princeton, NJ 2024-Present

- Developed and optimized GPU-accelerated scientific codes in collaboration with computational astrophysics research groups
- Extended Iseult, a Particle-in-Cell visualization code, adding key features and fixing long standing issues
- Ported the hydrodynamics code Ramses to GPUs as a proof of concept, achieving a 20-30x speedup
- Developed PegasusTools, a Python package for analyzing Pegasus++ PIC data, achieving 15-150x speedups
- Optimized the multiphysics astrophysical code Fornax, achieving a 1.8x speedup on the Aurora exascale supercomputer
- Ported the Athena++ astrochemistry module to the exascale GPU code AthenaK
- Contributed to the BDF ODE solvers in Kokkos Kernels

Ph.D. Candidate, University of Pittsburgh, Pittsburgh, PA 2018-2024

- Expanded [Cholla](#), a massively parallel GPU-accelerated code for simulating astrophysical fluid dynamics to include magnetic fields (magnetohydrodynamics/MHD) using state of the art methods.
- Conducted research with Professor Evan Schneider into numerical modeling of galactic winds using the GPGPU code [Cholla](#)
- Collaborated with the Frontier Center for Accelerated Application Readiness (CAAR) program to optimize Cholla to run on exascale supercomputers, namely Frontier.
- Established and executed a robust testing framework for Cholla, employing GoogleTest with custom extensions to ensure software reliability and quality.
- Led multiple initiatives to promote scientific software best practices within the Cholla development team, fostering excellence in software engineering standards.

Masters Student, Bowling Green, OH 2016-2018

- Conducted [original thesis research](#) into the properties of symbiotic star V1835 Aql with Professor Andrew Layden as advisor.
- Determined the general properties and causes of variability of a symbiotic star system through image and data analysis in Python.

Undergraduate Research Assistant, Purdue University, West Lafayette, IN 2015-2016

- Built an experimental two-channel dynamic digital holography system to investigate the time dependent effects of chemotherapy drugs on cancer tumors via biodynamic imaging.
- Improved efficiency & quality of large-scale (tens of terabytes) off-site data storage using HSI. Improved efficiency by a factor of 12.

FELLOWSHIPS AND AWARDS

Spot Award, Princeton University, 2025

Learning Beyond the Classroom Certificate, Purdue University, 2016

Presidential Scholarship, Purdue University, 2012-2016

Ascarelli Fellowship, Department of Physics and Astronomy, Purdue University, 2012

Eagle Scout, 2012

CONFERENCES AND PRESENTATIONS

- *Developing a Handbook of best practices for research software sustainability at Princeton.*, 4th Annual Conference of the US Research Software Engineer Association (US-RSE'26), October 2026. Sakuma, T.; Bretheim, J.; Cosden, I.; Murel, J.; **Caddy, R.**; et al.
 - *Getting Scientist buy-in on best practices: A Case Study.*, 2nd Annual Conference of the US Research Software Engineer Association (US-RSE'24), October 2024, Rapid Access Microtalk (RAM)
 - *Exascale MHD Simulations with Cholla*, Santa Cruz Organization for Outreach in Physics (SCOOP) No Jargon Talk, University of California Santa Cruz, *Invited Talk*, November 2023
 - *Exascale MHD Simulations with Cholla*, Seminar, NASA Goddard, *Invited Talk*, November 2023
 - *Exascale MHD Simulations with Cholla*, Center for Theory and Computation (CTC) Seminar, University of Maryland, *Invited Talk*, November 2023
 - *Exascale MHD Simulations with Cholla*, Joint Space Institute Workshop: Winds Throughout the Universe, *Contributed Talk*, October 2023
 - *GPU Accelerated Magnetohydrodynamics for Astrophysics & Testing for Exascale Codes*, Fall Meeting of the Mid-Atlantic Section of the APS, *Poster Presentation*, December 2022
 - *GPU Accelerated Magnetohydrodynamics for Astrophysics & Testing for Exascale Codes*, International High Performance Computing Summer School (IHPCSS), *Poster Presentation*, June 2022
 - *Time Series Photometry of the Symbiotic Binary V1835 Aql*, Ohio Academy of Sciences (OAS) Meeting, *Poster Presentation*, April 2018
 - *Time Series Photometry of the Symbiotic Binary NSV 11749*, Canadian-American-Mexican (CAM) Graduate Student Conference, *Poster Presentation*, August 2017
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PROFESSIONAL EXPERIENCE

Computational:

Languages: C++, C, Python, Fortran, Bash

Packages & API's: CUDA, HIP, OpenMP Offload, Kokkos, OpenMP, MPI, GoogleTest, Numpy, Polars

Software Tools: git, GitHub Actions, Clang Tools, $\LaTeX$, GCC, Make, HDF5, HSI, PBS/Slurm/LSF, DAOPHOT, IRAF, GoogleTest, Docker

HPC Resources Used: Supercomputer clusters at Purdue University, the University of Pittsburgh, Princeton University, Argonne National Lab Leadership Computing Facility Theta & Aurora supercomputers, Oak Ridge National Lab Leadership Computing Facility systems including Summit, Spock, Crusher, Andes, and Frontier.

Observing

PROMPT C1 & C5 at the Cerro Tololo Inter-American Observatory (*remote*) many nights observing the symbiotic star V1835 Aql

Service:

Member, Women and Minorities in Physics, University of Pittsburgh 2019-2024

President, Purdue Society of Physics Students (SPS), 2016

Member, Women in Physics, Purdue 2014-2016

APS - Conference for Undergraduate Women in Physics (Purdue University, 2015) - Volunteer

Professional Development:

- 2026 US Research Software Engineer Association 4th Annual Conference (US-RSE'26)
- 2025 US Research Software Engineer Association 3rd Annual Conference (US-RSE'25)
- 2024 Supercomputing Conference & Tutorials
- 2024 US Research Software Engineer Association 2nd Annual Conference (US-RSE'24)
- 2023 Supercomputing Conference & Tutorials
- 2023 US Research Software Engineer Association 1st Annual Conference (US-RSE'23)
- 2023 Practice and Experience in Advanced Research Computing Conference (PEARC)
- 2023 Platform for Advanced Scientific Computing Conference (PASC)
- 2022 Argonne Training Program for Extreme Scale Computing (ATPESC)
- 2022 International High Performance Computing Summer School (IHPCSS), Second place in the programming challenge
- 2021 Advanced Cyberinfrastructure Training for Modeling Physical Systems at Rensselaer Polytechnic Institute
- 2021 XSEDE Webinar: Performance Tuning and Single Processor Optimization
- 2020 Intel Developer Tools Workshop
- 2020 Frontier Center of Excellence (COE) Workshop, *invite only*
- 2020 OLCF User Meeting
- XSEDE HPC Workshop Series: Attended the MPI, OpenMP, Python & Performance, OpenACC, and Big Data & Machine Learning workshops

Membership: AAS Member, APS Member

SERVICE

Tutorials

- *Introduction to CI and Task Runners*, Princeton University, 2026
- *Introduction to CUDA*, Princeton University, 2025
- *Introduction to Python Data Types*, University of Pittsburgh, 2023
- *Introduction to Git, GitHub, and Git Workflows*, University of Pittsburgh, 2022
- *Scientific Software Best Practices*, University of Pittsburgh 2022, 2023
- *Organizing Your Dotfiles*, University of Pittsburgh, 2021

Mentorship

- Princeton Open Hackathon Mentor, Princeton University, 2026
- US-RSE'25 Conference Mentor, US Research Software Engineer Association, 2024
- Princeton Open Hackathon Mentor, Princeton University, 2025
- US-RSE'24 Conference Mentor, US Research Software Engineer Association, 2024

Teaching

- University of Pittsburgh, 2018-2020:
 - Graduate Teaching Assistant for introductory physics and astronomy courses
 - Led 2-5 recitations per week with 20-50 students each. Sometimes wrote my own recitation assignments/problems depending on what the professor required
 - Grading recitation assignments, exams, labs, etc.
 - Bowling Green State University, 2016-2018:
 - Graduate Teaching Assistant for introductory physics courses
 - Led 2-3 two hour labs per week of 20-30 students each
 - Purdue University, 2013-2016:
 - Undergraduate Teaching Assistant for introductory physics courses. Helped graduate TAs teach 2-4 recitations or labs per week of 20-30 students each
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SUBMITTED AND REFEREED PUBLICATIONS

★ - First or Second Author

4. *RAMSES-GPU: Cell-by-Cell Adaptive Mesh Refinement with Magneto-Hydrodynamics and Self-Gravity on Graphics Processing Units*. Teyssier R.; Moseley E.; **Caddy R.**; Haugbolle T.; Sunseri J.; Lee J.; Palafoutas J.; Pan Y.; Groger W.; Abel T.; Loring B.; Barnier G. 2026 DOI: [10.48550/arXiv.2607.20704](https://doi.org/10.48550/arXiv.2607.20704), arXiv:[2607.20704](https://arxiv.org/abs/2607.20704)
3. *Dynamical Dark Energy Imprints in the Lyman-Alpha Forest*
Garza, D.; Robertson, B.; Madau, P.; Gnedin, N.; Abbruzzo, M.W.; Schneider, E.; Budiardja, R.D.; White, J.B., III; **Caddy, R.**; Villaseñor, B., 2026, arXiv
DOI: [10.48550/arXiv.2601.00767](https://doi.org/10.48550/arXiv.2601.00767), arXiv:[2601.00767](https://arxiv.org/abs/2601.00767)
2. ★ *Cholla-MHD: An Exascale-capable Magnetohydrodynamic Extension to the Cholla Astrophysical Simulation Code*
Caddy, R. & Schneider, E., 2024, The Astrophysical Journal, Volume 970
DOI: [10.3847/1538-4357/ad464a](https://doi.org/10.3847/1538-4357/ad464a), arXiv:[2402.05240](https://arxiv.org/abs/2402.05240)
1. ★ *Optical Time-series Photometry of the Symbiotic Nova V1835 Aquilae*
Caddy, Robert V., Layden, Andrew C., Reichart, Daniel E., Haislip, Joshua B., Kouprianov, Vladimir V., Ivarsen, Kevin M., Moore, Justin P., LaCluyze, Aaron P., Linder, Tyler R., and Nysewander Melissa C., 2022 Publications of the Astronomical Society of the Pacific, Volume 134, Number 1039
DOI: [10.1088/1538-3873/ac8f6f](https://doi.org/10.1088/1538-3873/ac8f6f), arXiv:[2209.11251](https://arxiv.org/abs/2209.11251)